

MAC-2313 | CALCULUS III

Practice Exam 1

Original paper-style practice • no formula sheet

Name

Date

Score

Eight core problems, 10 points each. One optional bonus problem is worth up to 10 additional points. Show your reasoning: a correct method with a small arithmetic error may earn partial credit.

No formula sheet is provided. Assume the single-variable derivative and integral skills from Calculus I and substitution/integration by parts from Calculus II are available when useful. Exact answers are preferred unless an estimate is requested.

Suggested practice: attempt the exam without notes, then use the separate answer key to diagnose method choices and reasoning. No time limit is specified.

Problem 1 | Vectors and displacement

Show enough work for partial credit. State units and interpret your result when relevant.

A survey drone travels from $P=(-2,1,0)$ to $Q=(1,5,-2)$, then to $R=(4,2,3)$, with coordinates in meters.

(a) Find the displacement vector for each leg and the net displacement. (b) Find the total path length and the straight-line distance from P to R. Explain why they differ. (c) Give a unit vector pointing in the net-displacement direction. (d) A map reports only the net displacement. State one piece of route information it cannot recover.

Problem 2 | Dot products, projection, and work

Show enough work for partial credit. State units and interpret your result when relevant.

A constant force $F = \langle 6, 8, 0 \rangle$ N acts on a crate while it moves through displacement $d = \langle 3, 4, 12 \rangle$ m. Resolve the force relative to the motion.

(a) Find the work and explain its sign. (b) Find the scalar component of F along d and the vector projection of F onto d . (c) Determine $\cos(\theta)$, where θ is the angle between F and d . Explain why the projection magnitude is not the work.

Problem 3 | Cross products and plane equations

Show enough work for partial credit. State units and interpret your result when relevant.

Three noncollinear survey markers are $A=(1,2,0)$, $B=(3,1,1)$, and $C=(0,4,2)$. A fourth marker is $D=(2,2,4)$.

(a) Find a normal to the plane through A, B, C and derive a Cartesian equation. (b) Find the area of parallelogram ABC formed by AB and AC. (c) Decide whether D lies in the plane, showing a test. (d) Explain how reversing the order in the cross product changes the geometry and what it does not change.

Problem 4 | First and second partial derivatives

Show enough work for partial credit. State units and interpret your result when relevant.

Let $H(x,y)=2x^2y-xy^2+3x+5y$ model a local response surface. At $(1,2)$, determine H_x , H_y , H_{xx} , H_{xy} , H_{yx} , and H_{yy} . Then state the local first-order change estimate for a small move $(dx,dy)=(0.02,-0.01)$.

Problem 5 | Differentials and approximation

Show enough work for partial credit. State units and interpret your result when relevant.

A closed cylinder has total surface area $S(r,h)=2\pi r^2+2\pi rh$. At $r=3$ cm and $h=10$ cm, manufacturing changes the radius by $dr=0.02$ cm and height by $dh=-0.05$ cm.

(a) Estimate the change in surface area using the differential. (b) Estimate the relative percent change. (c) State whether the area rises or falls and explain why this is a local estimate rather than an exact change.

Problem 6 | Multivariable chain rule

Show enough work for partial credit. State units and interpret your result when relevant.

A response is $C(x,y)=x^2e^y+\sin(xy)$, where $x=1+t^2$ and $y=2-t$. Find dC/dt at $t=1$ using the multivariable chain rule. Show the point, partial derivatives, and inner derivatives. Check your result by differentiating $C(x(t),y(t))$ directly at $t=1$.

Problem 7 | Gradient and directional derivative

Show enough work for partial credit. State units and interpret your result when relevant.

An elevation model is $Z(x,y)=100-x^2-2xy-2y^2$. At $P=(1,-1)$, determine (a) the gradient, (b) the directional derivative toward $Q=(4,3)$, (c) the unit direction of steepest increase and its rate, and (d) the direction of steepest decrease. Interpret the directional derivative in context.

Problem 8 | Critical points and optimization

Show enough work for partial credit. State units and interpret your result when relevant.

A two-product shop models daily net profit by $P(x,y)=32x+36y-2x^2-2xy-3y^2$, where $x,y\geq 0$ are production levels in units of 100 items and P is dollars in hundreds.

(a) Find and classify the interior critical point using the second derivative test. (b) Explain why it is the unique global maximum on the feasible quadrant. (c) Compute and interpret the maximum modeled profit.

Optional bonus — variable-force work | 10 optional points

Show enough work for partial credit. State units and interpret your result when relevant.

A particle moves on the x -axis from $x=0$ m to $x=2$ m under force $F(x)= N$. Find the work. Explain why a dot product with the displacement vector alone is not enough when the force varies, and show the integration method.
